Response to Reviews: JCGS-25-420

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We thank the anonymous associate editor and reviewers for their careful reviews. What follows is our point-by-point response to reviewer comments.

Note that the original reviewer comments are in normal and *our response is in italic* text.

Reviewer 1

Summary:

This manuscript proposes an algorithm to evaluate nonlinear dimension reduction (NLDR) techniques. The authors identify some characteristics and limitations of NLDR and develop an algorithm to analyze NLDR results' quality of how well they match the original high-dimensional data.

Strengths:

1. This manuscript is well-organized, providing sufficient background knowledge and examples which makes this paper easy to follow.
2. This paper contains sufficient technical details in the introduction of the algorithm.

Weakness:

1. The main concern is whether the proposed algorithm can properly evaluate the NLDR's performance. Using the RMSE plot with different binwidth values (e.g., in Figure 8 and Figure 11) to evaluate NLDR's performance is problematic for not considering the density within each cluster. For NLDR results that have denser clusters, for example layout a,c,g,h in Figure 11, a smaller binwidth needs be used to achieve the same average count

in each bin, which will cause their lines in Figure 11 to be more upper left. However, denser clusters are not worse NLDR results.

Thank you for this thoughtful comment. We agree that the density of clusters can influence the RMSE curves, and that denser clusters are not inherently indicative of poorer NLDR results. The purpose of our diagnostic was not to penalize density per se, but to provide a way to compare the stability of neighborhood structures across different binwidths. In practice, a representation with highly dense clusters will indeed tend to have lower RMSE at small binwidths, shifting its curve to the upper-left of the plot. This is a reflection of how tightly the representation packs points together rather than a judgment of quality on its own.

To address this concern, we have revised the text to clarify that RMSE should not be interpreted in isolation. Instead, it should be considered in conjunction with other diagnostics, such as the proportion of non-empty bins and visual separation of clusters. We emphasize that our framework is intended to highlight trade-offs across multiple metrics rather than to define a single absolute ranking.

We also note in the revision that dense clusters are not inherently "worse" outcomes—rather, they may reflect meaningful biological or structural features of the data. The diagnostic is designed to make these differences visible so that users can interpret them in the context of their application.

2. In Figure 7, as RMSE steadily increases with the binwidth, it's not clear why the three binwidths 0.03, 0.05, and 0.07 are chosen. In addition, we shouldn't use a single simulated dataset to determine the default hyperparameters without any real analysis.

Thank you for raising this important point. We agree that in Figure 7, the steady increase of RMSE with binwidth makes it less obvious why specific values of a_1 *were selected. Our intent was not to claim that these three binwidths (0.03, 0.05, and 0.07) are universally optimal, but rather to use them as illustrative examples for exploring trade-offs between RMSE, bin density, and the proportion of non-empty bins. We chose these values because they represent a range of small-to-moderate binwidths that highlight changes across the diagnostic plots.*

We also agree with your concern that hyperparameters should not be set based on a single simulated dataset. To clarify, our goal here was to demonstrate the process of tuning rather than to prescribe default hyperparameters.

3. For several figures that contains results from multiple NLDR algorithms, it would be nice to add the information of which algorithm/ what hyperparameter values for each of the subfigure. Many readers may have tried several NLDR algorithms themselves, and they will benefit from those figures to gain insights about behaviors of different NLDRs. Also it's nice to mention which data is used to generate each figure (e.g., in Figure 7).

Thank you for this helpful suggestion. To improve the clarity and reproducibility of our results, we have now included a table summarizing the Figure, NLDR method, and the corresponding *hyper-parameters used for each. We believe this addition makes it easier for readers to interpret the visual results and to replicate or compare NLDR behavior across different parameter settings.*

4. In section 3.6.1, it says "Values of b1 between 2 and b1=sqrt(n/r2) are allowed", why the choice of b1 should consider r2?

These are the reasons:

- *b_1 controls the number of bins along the x-axis (the first NLDR dimension).*
- *In a hexagonal grid, the spacing along the y-axis (second dimension) is not independent. It's linked to the x-direction because hexagons "stack" with vertical offsets.*
- *The vertical extent is determined by the hexagon height, which in turn depends on the chosen bin width along the x-direction.*

So, to ensure the grid actually fits within the data range in both directions, the allowable values of b_1 are constrained by r_2, the range of the second dimension. If b_1 is too large, the hexagons become too narrow/tall, and the grid will not cover the vertical range properly.

That's why the condition is written as:

$$2 \leq b_1 \leq \sqrt{\frac{n}{r_2}}$$

tying the number of x-bins (b_1) to the vertical range of the data (r_2).

Reviewer 2

Summary

The manuscript proposes a novel, mostly graphical diagnostic to evaluate nonlinear dimension reduction (NLDR) layouts by effectively "lifting" the 2-D embedding back into the original high-dimensional space and inspecting the fit visually via tours (i.e., sequences of 2D-linear projections of the original high-D space). The goal is to help analysts discern which 2-D projection is the most "honest" (faithful) representation of the true structure in the data. The paper's contributions are:

1. an algorithm that overlays a geometric model on an NLDR layout using hexagonal binning and Delaunay triangulation in the embedding, then maps this model into the original large-D data space for visual assessment;

2. a new goodness-of-fit metric, the RMSE in original space, computed as the root mean squared distance between each point and its bin centroid in high-D (Eq 2) (3) interactive visualization tools (implemented in published software) to compare/investigate different NLDRs

It is mostly well written, adresses an important problem with critical real-world relevance, and provides a very well done, extensively documented R implementation. However, this reviewer is skeptical about the suitability of the newly proposed GOF metric to choose the most "reasonable" or "faithful" NLDR, finds the empirical evaluation of it to be severely lacking without comparison to alternative measures from the literature, and would request some restructuring and shortening of the paper.

Strengths

Relevance, Accessibility, and Intuitiveness

The way this method visualizes the spatial distortions induced by NLDR seems uniquely acces sible to me, and, coupled with the modern interactive/animated visualizations implemented for it, seems likely to become a very useful tool to investigate and understand high-D data structures while avoiding over-reliance on / misinterpretation of misleading 2D patterns so prevalent in many areas at the moment (e.g. Izarry (2024), Chari & Pachter (2023)).

Background Material & Examples

The paper is fully fleshed out with illuminating examples, and I commend the authors for the effort they put into documenting their package and providing very informative videos etc of the method in action.

Reproducibility and Code Availability

The paper is exemplary in this regard - the underlying software is a CRAN compliant R package, and the Github repository itself allows to recompile the entire paper and appendix with all figures etc from scratch.

Major Issues

No Standard Embedding Quality Metrics or Comparisons

My primary concern is the absence of established quantitative metrics

1. to evaluate to what extent the newly proposed “RMSE” metric corresponds to established notion of embedding faithfulness and
2. to validate the different embeddings under consideration.

The paper’s only evaluation of “honesty” is done via the proposed custom “RMSE” in high dimensional space (defined in Section 3.4) and visual heuristics, but it does not report any conventional measures like trustworthiness/continuity scores, k-Neighborhood preservation rates (as derived from the co-ranking matrix - e.g. R_{NX}-curves for different neighborhood sizes k, c.f. (Lee et al, 2015), (Lueks et al, 2011)), or Shepard diagrams of original space vs. embedding space distances. This is a significant omission because NLDR almost always faces a trade-off between optimizing preservation of local structures vs preservation of global structures, which these metrics are (mostly) designed to resolve / analyze. For example, the authors could have computed such measures of trustworthiness for each NLDR layout to see if the one with lowest “RMSE” indeed had the highest (local and or global) neighborhood structure preservation but no such analysis was provided. The manuscript doesn’t even mention or discuss these metrics, which is not acceptable in a study focused on faithful representation, in my opinion.

Questions a revised version should aim to resolve:

1. does “RMSE” measure local or global structure preservation (or a mixture of both)? From the definition, it seems to be focused on how well small neighborhoods in embed ding space (the bins)) correspond to small neighborhoods in high-D space, with, IIUC, particularly high values if there are lots of “hard intrusions” (i.e., points that are embed ded in the same 2D bin despite being far apart in high-D space)?
2. how strongly are “RMSE” values correlated to other established measures of (local or global) structure preservation? and (how) are these correlations affected by the binning hyperparameters?
3. are the NLDRs selected by lowest “RMSE” the ones that would also be selected by other, more established measures of embedding faithfulness whose characteristics are well understood? at the very least, such measures should be provided, compared and discussed for all the embeddings under consideration in the case studies.

The reliance of “RMSE” on euclidean distances in high-D space also seems limiting:

1. at least for coarser bins and/or highly non-linear data structures, geodesic distances along the data manifold à la ISOMAP are likely to measure a more relevant kind of dissimilarity
2. a definition based on a more general notion of distance would make this applicable to investigating 2D embeddings of data types for which the euclidean distance of their numerical data representations is not a suitable/applicable measure of dissimilarity (e.g. images, text).

Insufficient Validation of the Proposed Diagnostic (RMSE) Across Methods

Related to the above points - the paper has no theoretical or empirical demonstration that minimizing this "RMSE" aligns with minimizing other distortion measures like stress or maximizing neighborhoood preservation (as measured by the co-ranking matrix). Furthermore, the claim that RMSE can be used to compare different NLDR methods on equal footing de serves scrutiny. Different methods optimize different criteria; it's plausible that one method might (tends to) achieve lower "RMSE" simply by construction, especially if my reading that it mostly rewards embeddings that manage to avoid hard intrusions is correct. Overall, much more critical validation of the RMSE diagnostic is needed to convince readers that "honesty" as defined here aligns with meaningful goodness-of-fit in embedding.

Limitations of the Evaluation (Simulation Study Design):

The manuscript includes a simulation to illustrate the approach (the "2NC7" dataset with two nonlinear clusters in 7-D) and two real-data applications (single-cell RNA-seq and an image dataset of handwritten digits). While these examples are appropriate, the scope of simulation scenarios is narrow, and the analysis lacks any notion of variability. The simulation seems to be a single realization of a specific two-cluster scenario. The authors did not explore, for instance, varying the overlap between clusters, adding more clusters or different manifold shapes, or introducing different noise levels. Ideally, the authors would systematically vary factors like: number of clusters or pattern types, intrinsic dimensionality of structures, sample size, etc., and show that their approach reliably selects the truth (or at least, the most interpretable view) in each case, and how well alternative measures would do so. Because only one run is shown, there is also no information how stable the RMSE curves in Figures 8 and 13 are across replications. I encourage the authors to perform more systematic, replicated simulations, report measures of variability for their performance evaluations, compare against relevant alternatives, and take much more care not to overstate their conclusions beyond the tested scenarios (e.g. the claim that it "identify bad representations so they can be avoided" is based on a very limited set of examples).

Minor Issues

1. First part of the title seems inappropriate for a scientific paper to me, as does the implied (and not substantiated, IMO) claim that the proposal is (primarily) useful to reliably select one of many NLDRs (as opposed to primarily useful for visually investigating details about a specific 2D NLDR).
2. For marketing and clarity reasons, I would strongly suggest to not name the metric "RMSE", but something more distinctive/descriptive.

We agree that the name "RMSE" may cause confusion with the standard Root Mean Square Error metric. To improve clarity and better reflect the purpose of the measure, we have renamed it to RWBSS (Root Within-Bin Sum of Square) throughout the manuscript. This new name more accurately conveys that the metric quantifies structure preservation rather than prediction error.

3. Flow from 3.1 to 3.2 (with too many sub-subsections) is difficult. Sections 3.2.1 (Scale the data) and 3.2.2 (Construct hexagon grid) are just short paragraphs, would be clearer if combined into a single numbered list of steps for the model construction. Much of 3 might be clearer and more compact if simply replaced by a direct step-by-step description of the algorithm in algorithmic pseudocode form with suitable short explanations.

4. Section 3.5 is a claim without supporting evidence – many out-of-sample embedding methods exist (some based on rather similar ideas, which should be referenced), you would need to show that this particular one actually works reasonably well for this to deserve space in the paper. I would strongly suggest to simply mention this in the discussion section as a possible avenue for further development instead.

5. Section 3.6 should be drastically shortened and details moved into an appendix.

6. p. 18: "A particular pattern that we commonly see is that analysts tend to pick layouts with clusters that have big separations between them." The literature/internet contains many lively discussions of the perils of mis-interpreting and post-hoc'ing NL-DRs (e.g. (Izarry, 2024), (Chari & Pachter, 2023)), it might be good to cite/summarize these instead of/in addition to basing your motivation on personal anecdotal evidence? The tone here seems overly conversational to me (also: "we almost always see there are no big separations in the data") and should be made more formal.

7. Section 4, esp. p.19: IMO, "to compare and assess a range of representations" an analyst really should look at their respective RNX curves, stress values, and Shepard diagrams, think about whether they care more about local or global structure preservation, and then decide based on those factors. The method proposed here seems to me to be mostly useful to then investigate specific regions / slices of high-D space to see for which sets of data points the high-D structure is (not) preserved well in the NLDR. I find this whole section to be vastly overselling the proposal – it ignores and implicitly discards most of the prior work in this field. Its intro paragraphs are also highly redundant with other content in the paper and should be cut.

8. Section 5 should be moved into an appendix to shorten the paper somewhat. Figure in Section 5.1. could benefit from coloring the clusters in different colors so global structure preservation is also somewhat legible from the figures for the NLDRs (e.g. "is orange cluster between green and blue as in high-D or do they switch positions"). Section 5.2 might also have benefited from a corresponding ISOMAP or similar embedding of this data suitable for truthfully (isometrically!) unrolling the high-D manifold, and then showcasing that such a successful "unrolling" can be discerned from the way the

projections look? tSNE just isn't isometric, so the distortions visible here are entirely expected, IMO?

Typos and grammar:

Revised version should be much more carefully proofread: "appropraite", "doen't", "it is has two separated nonlinear clusters", "the methods tNSE"

We appreciate your careful attention to detail and for pointing out the typographical and grammatical errors. We have carefully proofread the entire manuscript and corrected the issues you identified (e.g., "appropraite" → "appropriate," "doen't" → "doesn't," "hastwo" → "has two," "tNSE" → "tSNE"), along with several other minor typographical inconsistencies. The revised version has been thoroughly checked to ensure clarity and correctness.

Loss function values

tSNE optimizes a Kullback–Leibler divergence between high- and low-dimensional pairwise similarities; however, computing the full loss exactly is feasible but expensive, and standard implementations often use approximations for efficiency. UMAP minimizes a cross-entropy loss over a fuzzy simplicial set, but the true loss cannot be fully tracked because stochastic gradient descent with limited negative sampling approximates the repulsive forces, making the exact loss both computationally prohibitive and noisy. PHATE does not explicitly calculate a conventional loss function; its embedding is derived from diffusion potential distances and MDS, so there is no meaningful per-epoch loss to report. TriMAP optimizes a loss based on triplet constraints, but the loss is generally only evaluated over sampled triplets rather than all possible triplets, meaning the full loss is not computed. PaCMAP, in contrast, reports an actual loss function during optimization, using a fixed set of negative samples for repulsion, making it possible to monitor per-epoch loss values reliably, though this estimate may be biased relative to the theoretical full pairwise objective.

Metrics

Neighborhood preservation metric ($R_{NX}(K)$)

Given:

- $X = x_1, \ldots, x_N$ — high-dimensional data
- $Y = y_1, \ldots, y_N$ — 2D (or low-dimensional) embedding

Step 1: Compute neighborhood ranks

For each point i:

- Compute pairwise distances in high-D: (_{ij} = | x_i - x_j |)
- Compute pairwise distances in low-D: (d_{ij} = | y_i - y_j |)

Then define the **rank** of (j) w.r.t. (i):

$$\rho_{ij} = \text{rank of } \delta_{ij} \text{ among all } j \neq i$$

$$r_{ij} = \text{rank of } d_{ij} \text{ among all } j \neq i$$

Step 2: Define K-nearest neighborhoods

$$\nu_i^K = j : 1 \leq \rho_{ij} \leq K$$
$$n_i^K = j : 1 \leq r_{ij} \leq K$$

Step 3: Compute overlap fraction for each point

$$Q_{NX}(K) = \frac{1}{NK} \sum_{i=1}^{N} |\nu_i^K \cap n_i^K|$$

Step 4: Rescale for comparability (Lee et al., 2015)

To adjust for random embedding baseline:

$$R_{NX}(K) = \frac{(N-1)Q_{NX}(K) - K}{N - 1 - K}$$

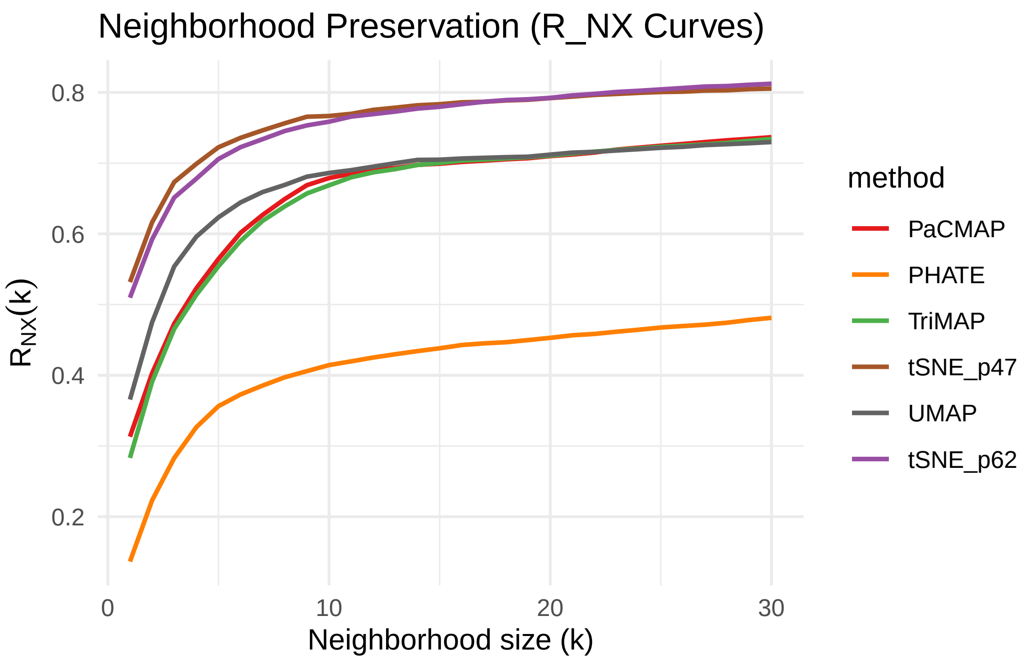
where (R_{NX}(K) [0,1]):

- (R_{NX}(K) = 1) → perfect preservation
- (R_{NX}(K) = 0) → random preservation

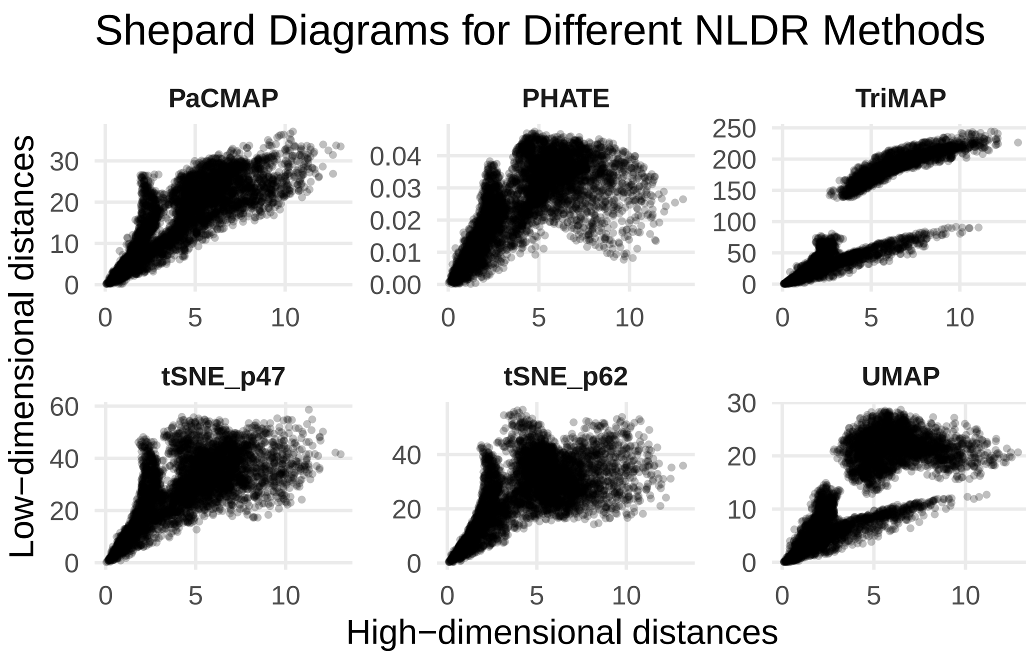
Step 5: Compute AUC over log-scaled (K)

$$\text{AUC} * \ln K(R * NX(K)) = \frac{\sum_{K=1}^{N/2} R_{NX}(K)/K}{\sum_{K=1}^{N/2} 1/K}$$

This gives a single scalar summarizing local–to–global preservation quality.



Shepard diagram



Random Triplet Accuracy (RTA)

To measure how well relative distances between points are preserved by a NLDR method without requiring class labels.

It checks how often the embedding preserves the correct ordering of pairwise distances from the high-dimensional space.

A **triplet** is a set of three points:

$$(i, j, k)$$

- i: anchor point
- j: a point close to i (in high-dimensional space)
- k: a point farther away from i

If in the high-dimensional space:

$$|x_i - x_j| < |x_i - x_k|$$

then the embedding Y should ideally preserve this ordering:

$$|y_i - y_j| < |y_i - y_k|$$

We define a *triplet as correctly preserved* if both inequalities hold. Formally, define an indicator:

$$I_{ijk} = \begin{cases} 1, & \text{if } (|x_i - x_j| < |x_i - x_k|) \text{ and } (|y_i - y_j| < |y_i - y_k|) \\ 0, & \text{otherwise} \end{cases}$$

Then, the **Random Triplet Accuracy (RTA)** is:

$$\text{RTA} = \frac{1}{|\mathcal{T}|} \sum_{(i,j,k) \in \mathcal{T}} I_{ijk}$$

where $\mathcal{T}$ is the set of sampled triplets.

Triplets can be sampled randomly across all points.

- For each anchor i, randomly sample a few pairs $(j, k) \neq i$.
- Use only distances between points to check ordering consistency.

(In current implementation, 5 random triplets per point, computes which ones preserve the ordering)

If labels are available, you can restrict j to be within the same class and k from a different class. This variant measures how well class structure is preserved.

Global score

Measures *how well a low-dimensional embedding* Y preserves the **global linear structure** of the high-dimensional data X.

We have:

- $X \in \mathbb{R}^{n \times p}$: the high-dimensional data (each row is an observation).
- $Y \in \mathbb{R}^{n \times d}$: the low-dimensional embedding (produced by t-SNE, UMAP, PHATE, etc.).
- n: number of observations, p: number of features, d: embedding dimension (usually 2).

The function computes a **global loss** based on how well Y can linearly reconstruct X.

We want to find a linear mapping $A \in \mathbb{R}^{p \times d}$ such that:

$$X^\top \approx AY^\top$$

Equivalently:

$$X \approx YA^\top$$

This means: can we **linearly reconstruct** the high-D data X from the low-D embedding Y? That's the key idea behind "global structure preservation": If Y preserves the global geometry of X, then there exists such a linear mapping A with low reconstruction error.

The optimal A (in least squares sense) minimizes:

$$\mathcal{L}(A) = |X^\top - AY^\top|_F^2$$

Setting the derivative $\frac{\partial \mathcal{L}}{\partial A} = 0$ gives:

$$A = X^\top Y(Y^\top Y)^{-1}$$

which matches exactly what the code computes (after transposing):

$$A = X^\top (Y(Y^\top Y)^{-1})$$

Now compute the **mean squared reconstruction error**:

$$L_{\text{global}}(X, Y) = \frac{1}{np}|X^\top - AY^\top|_F^2$$

This measures how well Y (through a linear transformation A) can reconstruct X.

- If Y retains global structure well; low L_{global}
- If Y distorts global structure; high L_{global}

Then, the function compares the embedding's global loss against that of a **PCA embedding**:

So the final **Global Score (GS)** is:

$$\text{GS} = \exp\left(-\frac{L_{\text{global}}(X, Y) - L_{\text{global}}(X, Y_{\text{PCA}})}{L_{\text{global}}(X, Y_{\text{PCA}})}\right).$$

So, the **Global Score** acts as a *normalized measure of how well the embedding preserves global linear relationships*, relative to PCA.

Method	Global_Score	Random_Triplet_Accuracy	R_NX_AUC	Shepard_Correlation
tSNE_p47	0.08	0.82	0.69	0.77

Method	Global_Score	Random_Triplet_Accuracy	R_NX_AUC	Shepard_Correlation
tSNE_p62	0.01	0.79	0.68	0.68
UMAP	0.06	0.81	0.58	0.80
PHATE	0.03	0.79	0.33	0.71
TriMAP	0.20	0.92	0.54	0.93
PaCMAP	0.14	0.87	0.55	0.84